Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Databricks Lakehouse Architecture, Delta Lake, Databricks Workflows, Unity Catalog, Spark Optimization, ETL/ELT Pipeline Development, Dimensional Modeling (Star/Snowflake Schemas, SCD Type 1/2), Batch & Streaming Data Workflows, Agentic AI Workflows, LLM Prompt Orchestration, MLOps, Autonomous Anomaly Detection, Time-Series Modeling (SARIMA), Data Governance & Lineage (Collibra)

Programming Languages & Tools: Databricks, Apache Spark, PySpark, Python, Structured Query Language (SQL), Azure (ADF, Databricks, Synapse, Azure SQL), AWS (S3, Redshift), Snowflake, Power BI (DAX), Tableau, Git, CI/CD Pipelines, KIRO IDE, Copilot

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Databricks Data & AI Engineer — **May 2022 – Present**

- **Architected high-throughput distributed pipelines and dimensional lakehouse models** using Databricks, Apache Spark, PySpark, Azure (ADF, Synapse, Azure SQL), and AWS (S3, Redshift) to crunch hundreds of terabytes with 99.9% data availability SLAs.
- **Engineered autonomous agentic AI workflows and prompt orchestration systems** using custom LLM tooling (KIRO IDE, Copilot) to automate schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50%.
- **Deployed an autonomous machine learning anomaly detection framework** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision using Python and SQL, safeguarding downstream enterprise analytics and pipeline integrity.
- **Implemented automated data quality firewalls and source-to-target reconciliation** integrated with Collibra data lineage to enforce strict data governance and data engineering best practices, reducing operational turnaround by over 80%.
- **Built and scaled batch and streaming data processing workflows** supporting complex analytics, conversational AI testing platforms (SpectrumIQ), and predictive ML use cases to modernize enterprise data foundations.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Developed predictive machine learning classification models** using XGBoost, Random Forests, and Scikit-Learn to analyze customer behavior and deliver actionable data signals, improving service reliability by 20%.
- **Architected advanced time-series forecasting models (SARIMA)** to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing response times by 20%.
- **Engineered automated testing and evaluation frameworks for enterprise virtual assistants** such as SpectrumIQ, cutting manual QA turnaround by 50% and accelerating production release cycles.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed and deployed executive Key Operating Metric (KOM) dashboards** and semantic models using Power BI (DAX) and Tableau to translate complex transactional telemetry into corporate strategy.
- **Bridged business and technology teams** by translating complex enterprise requirements into robust data architectures and scalable cloud analytics solutions.
- **Maintained meticulous attention to detail and data quality** across all reporting assets to ensure trusted executive decision-making and optimal platform performance.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.